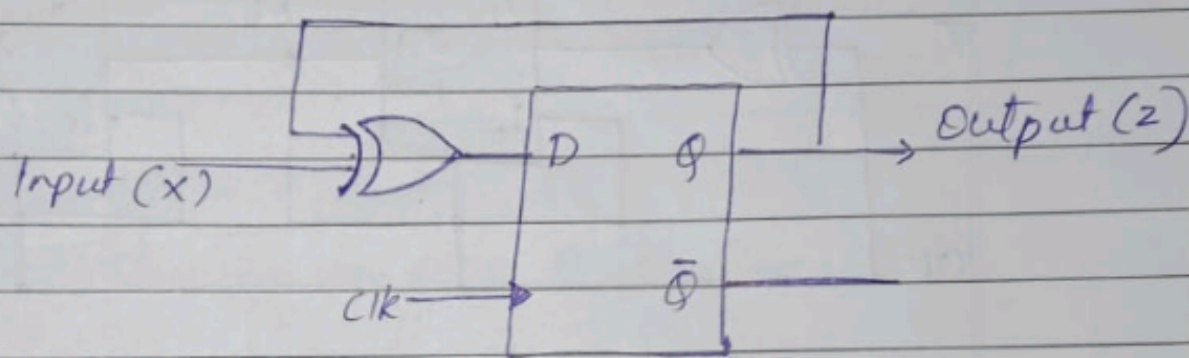


Assignment 4 - Sequential

1. Explain the behaviour of the given circuit.



$$\rightarrow D = X \oplus Q$$

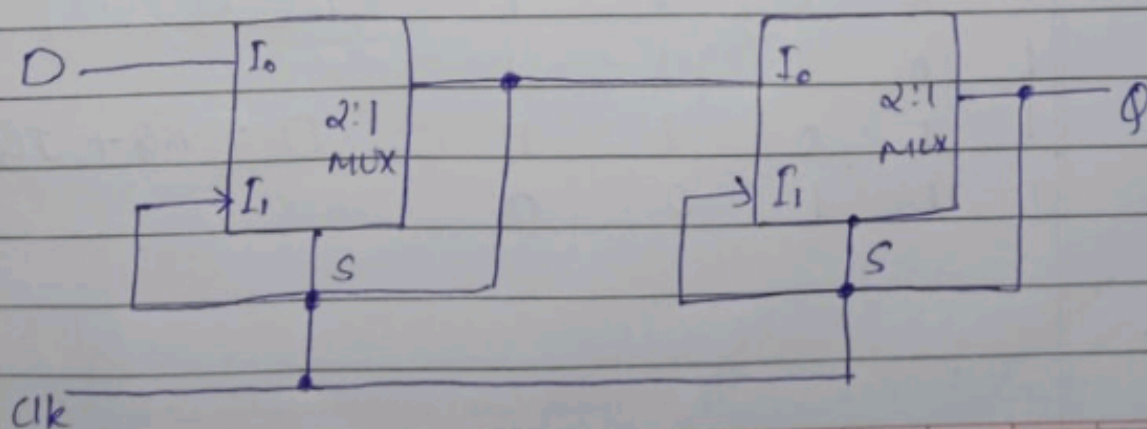
$$Q_{n+1} = X \oplus Q_n$$

If $X=0$: Output remains unchanged.

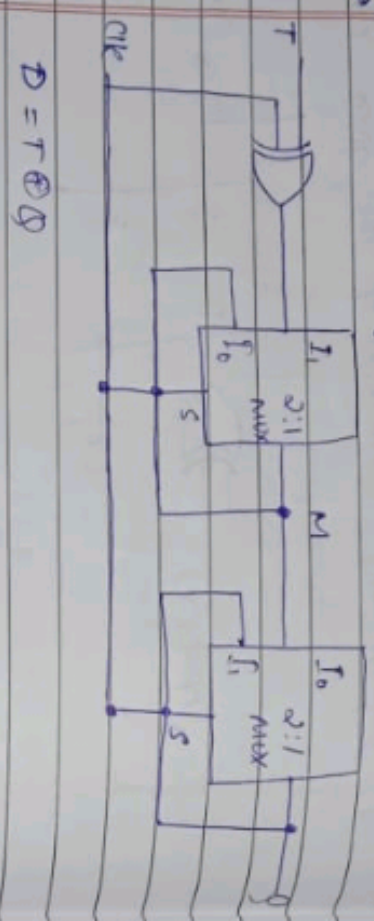
If $X=1$: Output toggles at every positive clock edge.

The given circuit behaves as a Positive edge triggered T Flip-Flop with $T = X$.

2. Design a posedge triggered D flipflop using 2:1 MUX.



3. Design a negedge triggered T flip-flop using 2:1 Muxes only.



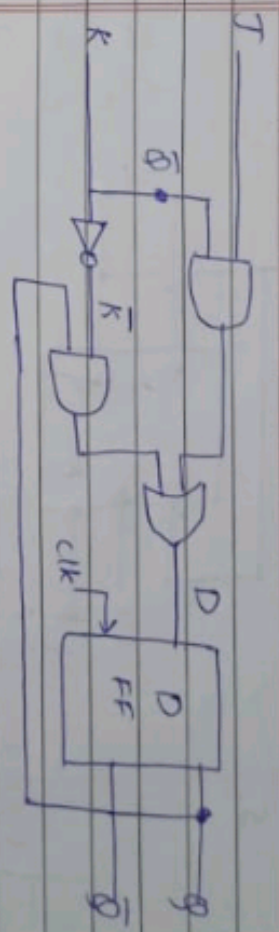
$$D = T \oplus Q$$

4. Convert by adding external gates:

a) A D flip-flop to a J-K flip-flop.

J	K	Q	Q'	D	
0	0	0	0	0	
0	0	1	1	1	
0	1	0	0	0	T
0	1	1	0	0	
1	0	0	1	1	KQ
1	0	1	1	1	
1	1	0	1	1	
1	1	1	0	0	

$$D = KQ + JQ'$$

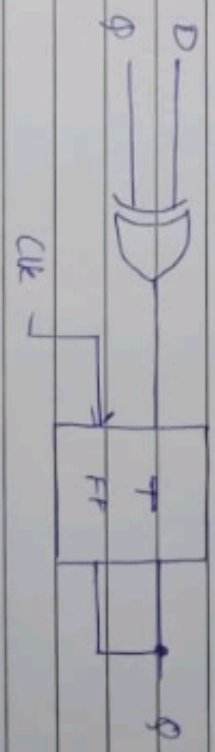


b) A T flip-flop to a D flip-flop

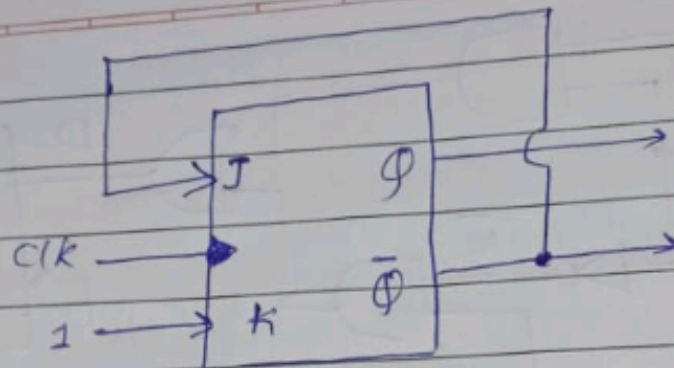
D	Q	Q'	T	
0	0	0	0	
0	1	0	1	
1	0	1	1	
1	1	1	0	

$$T = \overline{D}Q + DQ'$$

$$T = D \oplus Q$$



5.



<u>clk</u>	<u>Present ϕ</u>	<u>$J = \bar{\phi}$</u>	<u>K</u>	<u>Next ϕ</u>
1	0	1	1	1
2	1	0	1	0
3	0	1	1	1
4	1	0	1	0
5	0	1	1	1
6	1	0	1	0

$$\therefore \phi = 1, 0, 1, 0, 1, 0$$